

Redirections and Pipes in Bash - Terminal Style Summary

Definition:

Redirection changes the default input/output flow of commands, while pipes connect output of one command to the input of another, enabling command chaining.

Standard Streams in Bash

stdin (0) - Standard Input (keyboard)

stdout (1) - Standard Output (screen)

stderr (2) - Standard Error (screen)

1. Output Redirection (> and >>)

command > file # Overwrite file

command >> file # Append to file

Example:

```
echo "Hello World" > output.txt
```

```
ls >> filelist.txt
```

2. Input Redirection (<)

command < file

Example:

```
wc -l < myfile.txt
```

3. Error Redirection (2>, 2>>)

command 2> error.txt # Redirect errors

command 2>> error.txt # Append errors

Example:

```
ls /no_such_dir 2> errors.txt
```

4. Redirect stdout and stderr together

```
command > output.txt 2>&1
```

or

```
command &> output.txt
```

Example:

```
ls /etc /no_dir &> result.txt
```

5. Discard Output (/dev/null)

```
command > /dev/null 2>&1
```

This sends all output and errors to nowhere.

6. Here Document (<<)

```
cat << EOF
```

```
This is line one.
```

```
This is line two.
```

```
EOF
```

7. Here String (<<<)

```
grep "hello" <<< "hello world"
```

8. Combining Redirections

```
ls /etc /no_dir > output.txt 2> error.txt
```

Output and errors stored separately.

PIPES (|)

A pipe connects the output of one command to the input of another.

Syntax:

```
command1 | command2 | command3
```

Examples:

```
ls -l | grep ".sh"           # Show only shell scripts
```

```
ps aux | wc -l               # Count processes
```

```
cat /etc/passwd | sort      # Sort users
cat names.txt | sort | uniq # Remove duplicates
cat /var/log/syslog | grep "error" > errors.txt
```

Quick Reference Table

Operator	Description	Example
>	Redirect output (overwrite)	echo hi > file.txt
>>	Redirect output (append)	echo hi >> file.txt
<	Redirect input	wc -l < file.txt
2>	Redirect errors	ls /no_dir 2> err.txt
&>	Redirect all output	cmd &> all.txt
	Pipe output	ls grep txt
<<	Here Document	cat << EOF
<<<	Here String	grep word <<< "text"